



EC-R3576PC

| Low-power Large-model Computer

V1.1 2026-5-7

T-CHIP INTELLIGENCE TECHNOLOGY



Product features



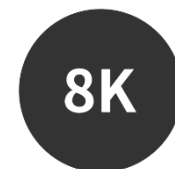
High-performance octa-core 64-bit AIOT processor, RK3576

RK3576, the new octa-core 64-bit AIOT processor, features a big.LITTLE architecture (4×A72 +4×A53), an advanced lithography process, and a frequency of up to 2.2 GHz.



Built-in 6TOPS powerful computing power NPU

Supports mixed operations such as INT4/INT8/INT16, and can achieve network model transformation based on TensorFlow/MXNet/PyTorch and other frameworks.



8K@30fps Decoding/4K@60fps Encoding

It supports 8K@30fps/4K@120fps decoding(H.265/HEVC, VP9, AVS2, AV1) , 4K@60fps decoding(H.264/AVC), 4K@60fps encoding(H.265/HEVC, H.264/AVC).



Strong network communication capabilities

It has 1 × Gigabit Ethernet, 1 × 100M Ethernet, supports 2.4GHz/5GHz dual-band WiFi, Bluetooth 5.0, and can expand the 4G LTE mobile network, so that network communication has a higher speed.

Product features



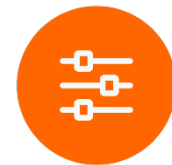
Various operating systems

Support Android14, Linux OS and Buildroot+QT. These provide safe and stable systems for product research and production.



Industrial-grade metal casing for efficient heat dissipation

Equipped with an industrial-grade all-metal casing and a built-in fan, it features efficient active heat dissipation, low power consumption and low noise, supporting 7×24 hours of uninterrupted and stable operation.



Abundant expansion interfaces

Equipped with HDMI2.1, Gigabit Ethernet, USB3.0, USB2.0, Type-C, RS232, RS485, CAN and other extension interfaces to meet the needs of different scenarios.



A wide range of applications

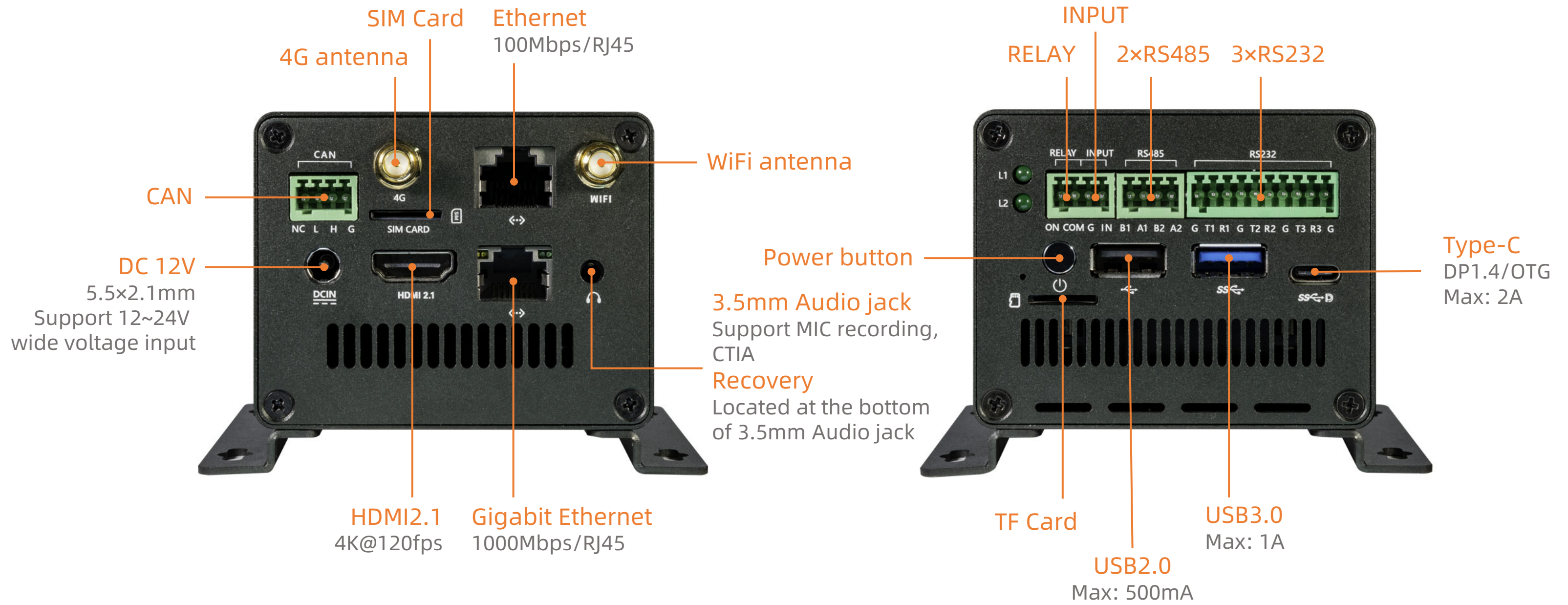
It is widely used in edge computing, local deployment of large models, intelligent digital signage, cloud terminal products, industrial PCs, automotive electronics, and more.



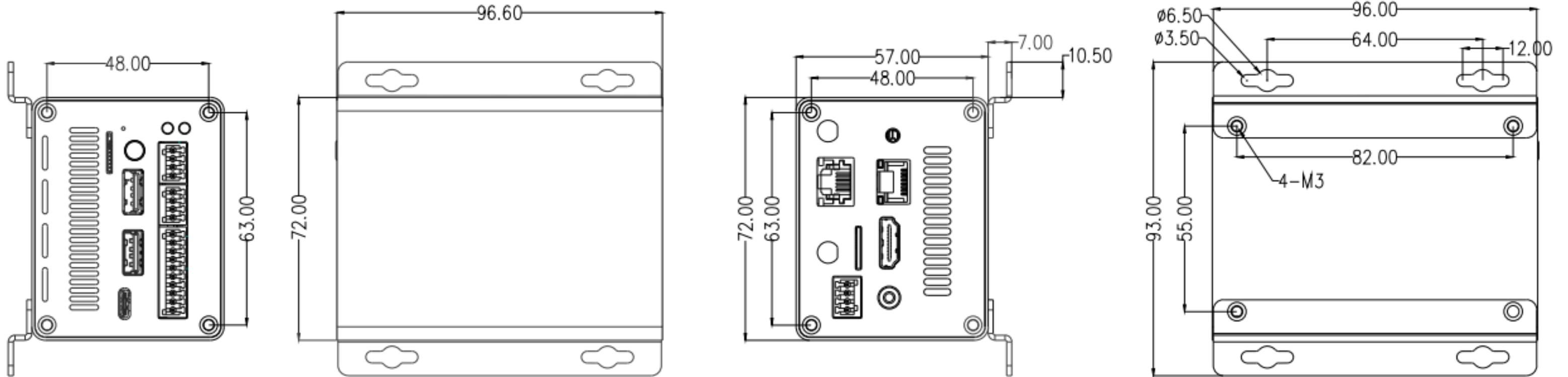
Specifications

Specifications		
Basic Specifications	SOC	Rockchip RK3576
	CPU	Octa-core 64-bit processor (4xA72 + 4xA53), up to 2.2GHz
	GPU	G52 MC3 @ 1GHz, support OpenGL ES 1.1/2.0/3.2, OpenCL 2.0, Vulkan 1.1, with high-performance 2D acceleration hardware embedded
	NPU	6 TOPS NPU, support mixed operations of INT4/8/16/FP16/BF16/TF32
	Decode/Encode	Decode: 8K@30fps/4K@120fps: H.265/HEVC, VP9, AVS2, AV1, 4K@60fps: H.264/AVC Encode: 4K@60fps: H.265/HEVC, H.264/AVC
	RAM	LPDDR4/LPDDR4x/LPDDR5 (4GB/8GB/16GB optional)
	Storage	eMMC (16GB/32GB/64GB/128GB/256GB optional), UFS3.1 (optional)
	Storage Expansion	1 × M.2 (Scalable 2242 PCIe NVMe/SATA SSD) (Inside the Computer), 1 × TF Card
	Power	DC 12V (5.5mm × 2.1mm, support 12V~24V wide voltage input)
	Power consumption	Min: 0.024W(12V/2mA), Normal: 4.8W(12V/400mA), Max: 10.8W(12V/900mA)
	OS	Android14, Linux OS, Buildroot+QT
	Software support	• Support the privatization deployment of ultra-large-scale parametric models under the Transformer architecture, such as Gemma-2B, ChatGLM3-6B, Qwen-1.8B, Phi-3-3.8B and other large language models • It supports traditional network architectures such as CNN, RNN, and LSTM, and supports the import and export of RKNN models; Support a variety of deep learning frameworks, including TensorFlow, TensorFlow Lite, PyTorch, Caffe, ONNX and Darknet. It also supports the development of custom operators • Support Docker container management technology
	Size	96.6mm × 93.0mm × 64.0mm
	Weight	≈0.45kg
	Environment	Operating Temperature: -20°C ~ 60°C, Storage Temperature: -20°C ~ 70°C, Storage Humidity: 10% ~ 90%RH (non-condensing)
Interface Specifications	Internet	1 × Gigabit Ethernet (RJ45), 1 × 100Mbps Ethernet(RJ45), support 2.4GHz/5GHz dual-band WiFi (802.11a/b/g/n/ac), Bluetooth 5.0, expandable 4G LTE
	Video output	1 × HDMI2.1 (4K@120fps), 1 × DP1.4 (4K@120fps), support dual-screen display
	Watchdogs	External watchdogs
	USB	1 × USB3.0 (Max:1A), 1 × USB2.0 (Max: 500mA), 1 × Type-C (OTG/DP1.4, Max: 2A)
	Expansion interface	3 × RS232, 2 × RS485, CAN, RELAY, INPUT, 1 × 3.5mm Audio jack (Support MIC recording, American Standard CTIA) , 1 × Recovery, 1 × SIM Card

Interface description



Dimension





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